

Question 1

(a) Given $r(x, y) = \sqrt{x^2 + y^2} = (x^2 + y^2)^{\frac{1}{2}}$

$$\begin{aligned}r_x &= \frac{1}{2}(x^2 + y^2)^{-\frac{1}{2}} \cdot 2x \\&= x(x^2 + y^2)^{-\frac{1}{2}} \\r_{xx} &= ()\end{aligned}$$

Question 2**Question 3**